



SYMBIOSIS INSTITUTE OF TECHNOLOGY, NAGPUR

Symbiosis International (Deemed University)

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Practical No. : 4

Aim –

Use JavaScript function types, scope, closures; apply try-catch; build a palindrome checker.

Software/Tools Required –

VSCode, Chrome, HTML5, CSS3, Javascript (ES6)

Experiment Program Code –

index.html

```
<!DOCTYPE html>
<html lang="en">
<head>
<meta charset="UTF-8">
<meta name="viewport" content="width=device-width, initial-scale=1.0">
<title>Palindrome Checker</title>
<link href="https://fonts.googleapis.com/css2?family=Poppins:wght@300;400;500;600&display=swap"
rel="stylesheet">

<style>
*{
margin:0;
padding:0;
box-sizing:border-box;
font-family:'Poppins',sans-serif;
}

body{
height:100vh;
display:flex;
justify-content:center;
align-items:center;
overflow:hidden;
background:linear-gradient(-45deg,#ff9a9e,#fad0c4,#a18cd1,#89f7fe);
background-size:400% 400%;
animation:bg 12s ease infinite;
}

@keyframes bg{
0%{background-position:0% 50%;}
50%{background-position:100% 50%;}
100%{background-position:0% 50%;}
}
```

```
.circle{
position:absolute;
border-radius:50%;
background:rgba(255,255,255,.25);
backdrop-filter:blur(5px);
animation:float 8s infinite ease-in-out;
}

.c1{
width:220px;
height:220px;
top:-60px;
left:-60px;
}

.c2{
width:160px;
height:160px;
bottom:-40px;
right:60px;
animation-delay:2s;
}

.c3{
width:120px;
height:120px;
top:65%;
left:18%;
animation-delay:4s;
}

@keyframes float{
50%{
transform:translateY(-25px);
}
}

.container{
width:430px;
padding:40px;
background:rgba(255,255,255,.20);
backdrop-filter:blur(18px);
border-radius:30px;
border:1px solid rgba(255,255,255,.4);
box-shadow:0 20px 50px rgba(0,0,0,.15);
text-align:center;
}

.icon{
width:90px;
height:90px;
margin:auto;
margin-bottom:20px;
border-radius:50%;
display:flex;
justify-content:center;
align-items:center;
font-size:42px;
background:white;
box-shadow:0 10px 25px rgba(0,0,0,.1);
}

h1{
color:white;
margin-bottom:8px;
}
```

```

p{
color:white;
opacity:.95;
margin-bottom:25px;
}

input{
width:100%;
padding:15px;
border:none;
border-radius:15px;
outline:none;
font-size:16px;
margin-bottom:20px;
transition:.3s;
}

input:focus{
transform:scale(1.02);
box-shadow:0 0 18px rgba(255,255,255,.8);
}

button{
width:100%;
padding:15px;
border:none;
border-radius:15px;
background:linear-gradient(90deg,#ff758c,#ff7eb3);
color:white;
font-size:17px;
font-weight:600;
cursor:pointer;
transition:.3s;
}

button:hover{
transform:translateY(-4px);
box-shadow:0 15px 30px rgba(255,118,163,.45);
}

.footer{
margin-top:22px;
color:white;
font-size:13px;
opacity:.8;
}
</style>
</head>

<body>

<div class="circle c1"></div>
<div class="circle c2"></div>
<div class="circle c3"></div>

<div class="container">
<div class="icon">❏</div>

<h1>Palindrome Checker</h1>

<p>Check whether your word reads the same forwards and backwards.</p>

<input
type="text"
id="word"
placeholder="Enter a word...">

```

```

<button onclick="checkPalindrome()">
✳️ Check Word
</button>

<div class="footer">
Developed by Sanskruti Gorle | 24070521025
</div>
</div>

<script>
function reverseString(str){
return str.split("").reverse().join("");
}

function message(){
var msg="Palindrome Checker";

function display(){
return msg;
}

return display;
}

function checkPalindrome(){
try{
var word=document.getElementById("word").value.trim();

if(word==""){
throw "Please enter a word.";
}

if(!/^[A-Za-z]+$/.test(word)){
throw "Only alphabets are allowed.";
}

if(word.length<3){
throw "Please enter at least 3 characters.";
}

if(word.length>20){
throw "Maximum 20 characters are allowed.";
}

let input=word.toLowerCase();
var reverse=reverseString(input);

document.write(
<!DOCTYPE html>
<html>
<head>
<title>Result</title>

<link          href="https://fonts.googleapis.com/css2?family=Poppins:wght@300;400;500;600&display=swap"
rel="stylesheet">

<style>
*{
margin:0;
padding:0;
box-sizing:border-box;
font-family:'Poppins',sans-serif;
}

body{
height:100vh;
display:flex;

```

```
justify-content:center;
align-items:center;
background:linear-gradient(-45deg,#ff9a9e,#fad0c4,#a18cd1,#89f7fe);
background-size:400% 400%;
animation:bg 12s ease infinite;
}
```

```
@keyframes bg{
0%{background-position:0% 50%;}
50%{background-position:100% 50%;}
100%{background-position:0% 50%;}
}
```

```
.card{
width:440px;
padding:40px;
background:rgba(255,255,255,.92);
border-radius:30px;
box-shadow:0 20px 50px rgba(0,0,0,.15);
text-align:center;
}
```

```
.icon{
font-size:60px;
margin-bottom:15px;
}
```

```
h2{
color:#444;
}
```

```
hr{
margin:20px 0;
border:none;
height:2px;
background:#eee;
}
```

```
.word{
font-size:20px;
margin-bottom:20px;
color:#555;
}
```

```
.success{
color:#16a34a;
font-size:24px;
font-weight:600;
}
```

```
.fail{
color:#ef4444;
font-size:24px;
font-weight:600;
}
```

```
button{
margin-top:30px;
padding:14px 28px;
border:none;
border-radius:15px;
background:linear-gradient(90deg,#ff758c,#ff7eb3);
color:white;
font-size:16px;
cursor:pointer;
transition:.3s;
}
```

```

button:hover{
transform:scale(1.05);
}
</style>
</head>

<body>

<div class="card">
<div class="icon">${input==reverse ? "🐼" : "🐻"}</div>

<h2>${message()}</h2>

<hr>

<div class="word">
<b>Entered Word:</b> ${word}
</div>
`);

if(input==reverse){
document.write(`
<div class="success">
🐻 It's a Palindrome!
</div>
`);
}
else{
document.write(`
<div class="fail">
🐼 Not a Palindrome
</div>
`);
}

document.write(`
<button onclick="history.back()">
🔍 Check Another Word
</button>

</div>

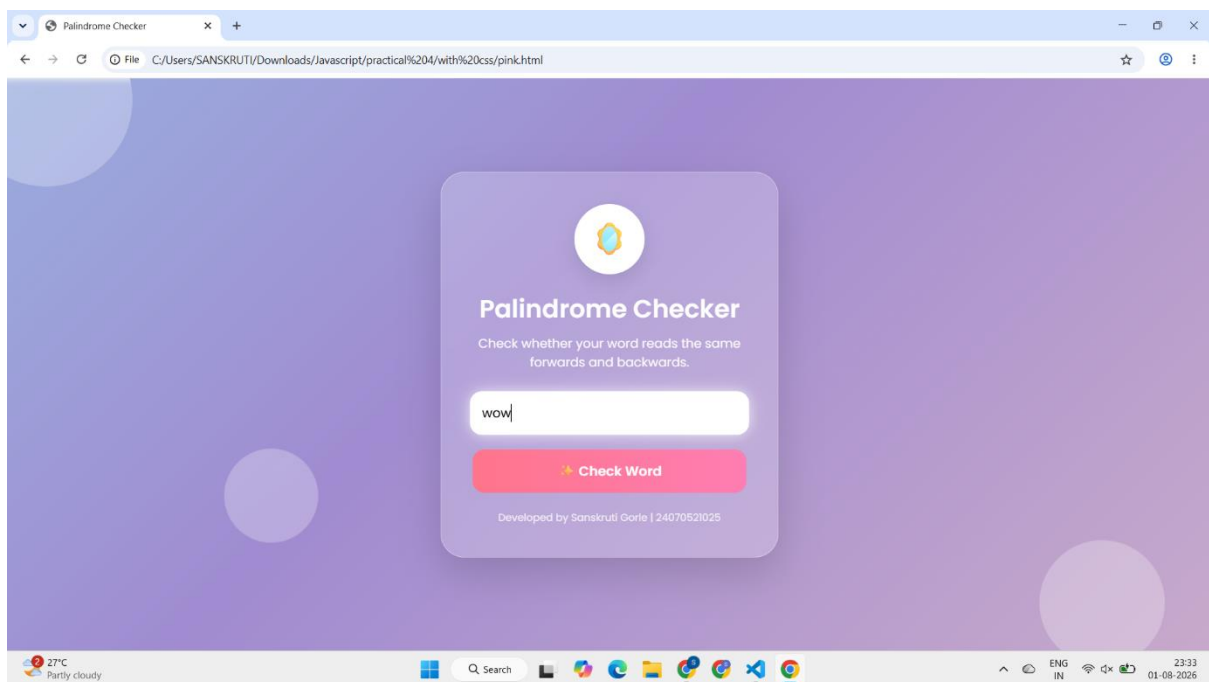
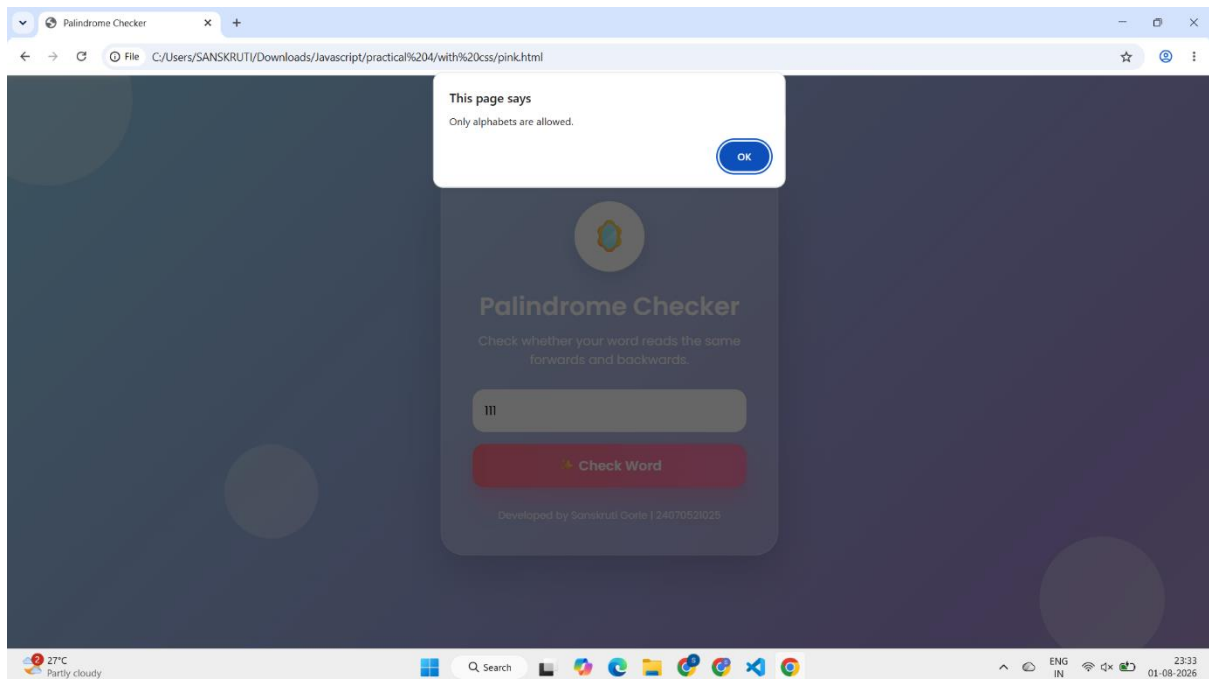
</body>
</html>
`);
}

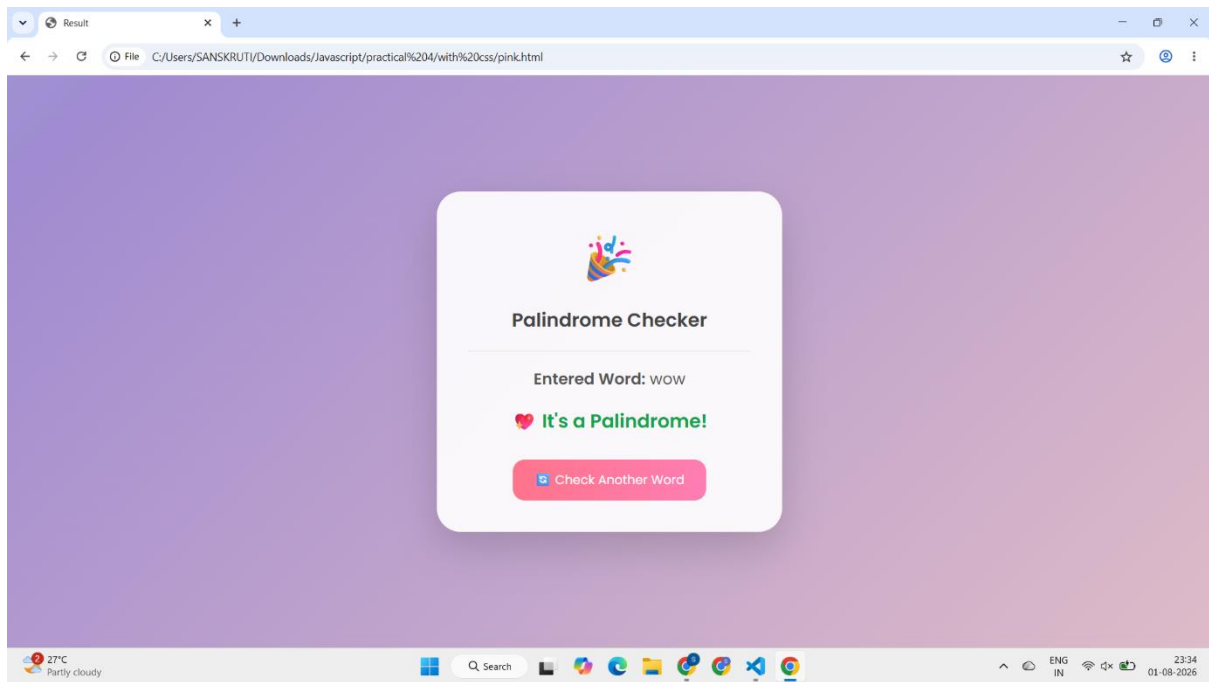
catch(error){
alert(error);
}
}
</script>

</body>
</html>

```

Output –





Case Study Title –

To develop a web application using HTML and JavaScript that validates a vehicle registration number using JavaScript functions and exception handling (try-catch) based on predefined validation rules, and displays whether the registration number is valid or invalid.

Case Study Program Code –

index.html

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>Vehicle Registration Validator</title>

  <link rel="stylesheet" href="style.css">

  <link rel="preconnect" href="https://fonts.googleapis.com">
  <link rel="preconnect" href="https://fonts.gstatic.com" crossorigin>

  <link href="https://fonts.googleapis.com/css2?family=Poppins:wght@300;400;500;600;700&display=swap"
rel="stylesheet">

  <link rel="stylesheet"
href="https://cdn.jsdelivr.net/npm/font-awesome@6.6.0/css/all.min.css">
</head>
<body>
```



```

<div class="container">

  <div class="icon">
    
  </div>

  <h1>Vehicle Registration</h1>

  <p class="subtitle">
    Verify your vehicle registration number
  </p>

  <label>Registration Number</label>

  <input
    type="text"
    id="regNo"
    maxlength="10"
    placeholder="Ex: MH12AB1234"
  >

  <button onclick="validateRegistration()">
    Validate Registration
  </button>

  <div id="result"></div>

  <div class="credits">
    <p>Developed by</p>
    <strong>Sanskriti Gorle</strong>
    <span>PRN: 24070521025</span>
  </div>

</div>

<script src="script.js"></script>
</body>
</html>

```

style.css

```

*{
  margin:0;
  padding:0;
  box-sizing:border-box;
  font-family:'Poppins',sans-serif;
}

body{

  display:flex;
  justify-content:center;
  align-items:center;

  min-height:100vh;

  background:#F6F3EE;

  padding:20px;
}

.container{

```

```
width:430px;

background:#FFFFFF;

padding:40px;

border-radius:24px;

box-shadow:
  0 15px 40px rgba(0,0,0,.08),
  0 3px 8px rgba(0,0,0,.04);

transition:.35s;
}

.container:hover{

  transform:translateY(-4px);

  box-shadow:
    0 22px 45px rgba(0,0,0,.10),
    0 5px 12px rgba(0,0,0,.05);
}

.icon{

  width:80px;
  height:80px;

  margin:auto;

  background:#E8F1E8;

  border-radius:50%;

  display:flex;
  justify-content:center;
  align-items:center;

  font-size:38px;

  margin-bottom:20px;
}

h1{

  text-align:center;

  color:#3E3E3E;

  font-size:30px;

  margin-bottom:10px;
}

.subtitle{

  text-align:center;

  color:#777;

  margin-bottom:35px;

  font-size:15px;
}
```

```
label{

    display:block;

    margin-bottom:10px;

    color:#555;

    font-weight:500;
}

input{

    width:100%;

    padding:16px;

    border:2px solid #E4E4E4;

    border-radius:14px;

    font-size:17px;

    outline:none;

    transition:.3s;

    text-transform:uppercase;
}

input:focus{

    border-color:#7B9E7B;

    box-shadow:0 0 5px rgba(123,158,123,.15);
}

button{

    width:100%;

    margin-top:24px;

    padding:16px;

    border:none;

    border-radius:14px;

    background:#6D8B74;

    color:white;

    font-size:17px;

    font-weight:600;

    cursor:pointer;

    transition:.3s;
}

button:hover{

    background:#56735D;
```

```
}

button:active{

    transform:scale(.98);

}

#result{

    margin-top:28px;

    border-radius:14px;

    padding:18px;

    display:none;

    animation:fade .35s ease;

}

#result h3{

    margin-bottom:8px;

}

#result p{

    line-height:1.5;

}

.credits{

    margin-top:30px;

    padding-top:18px;

    border-top:1px solid #EAEAEA;

    text-align:center;

}

.credits p{

    font-size:13px;

    color:#888;

    margin-bottom:4px;

}

.credits strong{

    display:block;

    color:#444;

    font-size:15px;

    font-weight:600;

}

.credits span{

    display:block;

    margin-top:3px;

    color:#888;

    font-size:12px;

}

@keyframes fade{

    from{

        opacity:0;

        transform:translateY(12px);

    }

}
```

```

    }

    to{

        opacity:1;
        transform:translateY(0);

    }

}

```

script.js

```

function validateRegistration() {

    let reg = document.getElementById("regNo").value.toUpperCase().trim();
    let result = document.getElementById("result");

    try {

        if (reg == "") {
            throw "Registration number cannot be empty.";
        }

        if (reg.length != 10) {
            throw "Registration number must contain exactly 10 characters.";
        }

        if (!/^[A-Z]{2}/.test(reg)) {
            throw "First two characters must be uppercase alphabets (State Code).";
        }

        if (!/^[A-Z]{2}[0-9]{2}/.test(reg)) {
            throw "Next two characters must be digits (District Code).";
        }

        if (!/^[A-Z]{2}[0-9]{2}[A-Z]{2}/.test(reg)) {
            throw "Next two characters must be uppercase alphabets (Series).";
        }

        if (!/^[A-Z]{2}[0-9]{2}[A-Z]{2}[0-9]{4}$/.test(reg)) {
            throw "Last four characters must be digits (Vehicle Number).";
        }

        result.style.display = "block";

        result.style.background = "#EDF8EE";
        result.style.border = "2px solid #8BC48A";
        result.style.color = "#295C2F";

        result.innerHTML = `
        <h3>✔ Registration Verified</h3>
        <p>
        This vehicle registration number is valid.
        </p>
        `;

    }

    catch (error) {

        result.style.display = "block";

        result.style.background = "#FFF0F0";
    }
}

```

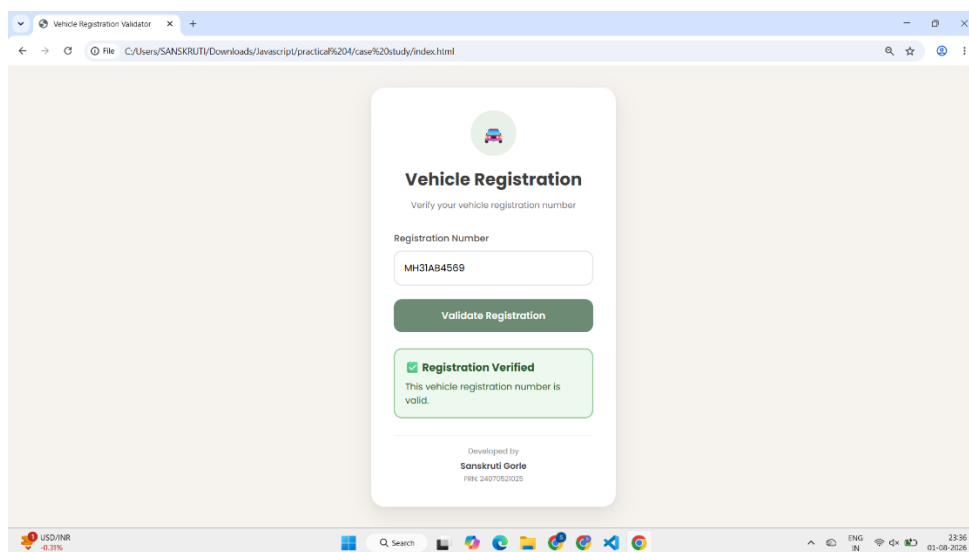
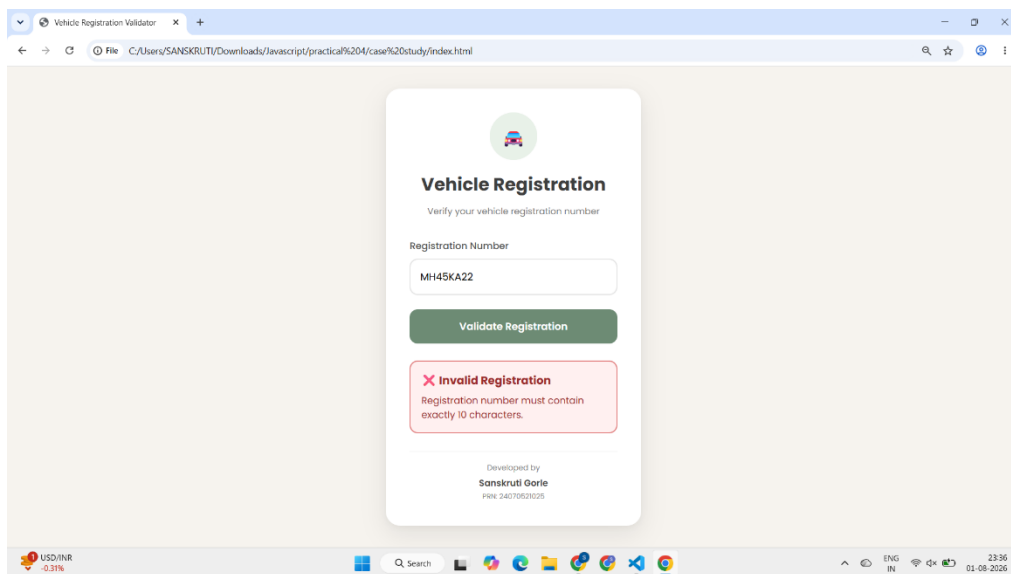
```

result.style.border = "2px solid #E27D7D";
result.style.color = "#9B2C2C";

result.innerHTML = `
<h3>✖ Invalid Registration</h3>
<p>${error}</p>
`;
}
}

```

Output –



Conclusion –

The experiment was successfully performed to understand JS functions, variable scope, closures, and exception handling using try-catch. A Vehicle Registration Validator and a Palindrome Checker were developed, these applications demonstrated input validation, string manipulation, and error handling using JavaScript.